Advanced Biomedical Artificial Intelligence

BME 6938, Section TBD

Class Periods: Tuesdays 1:55 PM-2:45 PM (Period 7) and Thursdays 1:55 PM-3:50 PM (Periods 7-8)

Location: BLK 0315

Academic Term: Spring 2027

Instructor:
Dr. Parisa Rashidi
Email: TBA
(352)-392-9469
Office Hours: TBA

Course Description

(3 credit hours) Examines current research in biomedical artificial intelligence for graduate students who have completed an introductory machine learning or deep learning course. Topics include self-supervised representation learning; foundation models for medical imaging, electronic health records, and physiological time series; vision-language and multimodal models; adaptation of large language models; agentic systems; generative models; clinical evaluation; robustness and fairness; and deployment. The course combines a weekly lecture, student-led paper sessions, and a team term project.

Course Pre-Requisites / Co-Requisites

BME 6938 Medical Artificial Intelligence (Fundamentals of Biomedical AI) or an equivalent machine learning or deep learning course, and proficiency in Python. Familiarity with neural network training and the transformer architecture is expected. A transformer refresher is provided in Week 1.

Course Objectives

- Explain the principles behind self-supervised pretraining, foundation models, and multimodal and agentic architectures used in biomedicine.
- Critically read, present, and critique current biomedical AI research papers.
- Formulate an open problem in biomedical AI and select appropriate data, models, and adaptation methods to address it.
- Evaluate models beyond accuracy, including calibration, subgroup performance, and robustness to distribution shift.
- Identify the requirements for translating AI models into clinical use, including prospective evaluation, monitoring, and human-AI interaction.
- Design, carry out, and report a team research project in conference-paper format.

Course Format

Tuesday lecture (Period 7): An instructor mini-lecture on the core concept of the week, a curated video such as a conference keynote or author talk, and a preview of the papers for Thursday.
Thursday paper sessions (Periods 7-8): Two papers per week, one per period, typically one methods paper and one biomedical application paper. Each session has a presentation group (20 min), a critique group (15 min), and class discussion (15 min). Each group presents about twice and critiques about twice during the term.
Reading responses: Students who are not presenting or critiquing submit a one-paragraph response and one discussion question for each paper before the Thursday class. Responses are graded complete or incomplete.

Term Project

Teams of three carry out an original study on an open problem in biomedical AI of their own choosing and report it as a research paper. Topics are approved at the proposal stage. Teams that do not have a topic may choose from a list of suggested problems provided by the instructor.

Evaluation standard: every paper must include a comparison against an appropriate baseline, an evaluation that goes beyond a single aggregate metric (for example calibration, subgroup performance, or a distribution-shift or robustness test, as appropriate to the problem), and a failure analysis.
Deliverables: a one-page proposal stating the open problem, data, methods, and evaluation plan; a two-minute pitch with a written peer review of another team's proposal; a one-page progress report with baseline results; a final presentation (9 min plus 3 min of questions); a 4-page conference-style paper; and a reproducible GitHub repository.
Data and computing: Datasets that require credentialed access (for example MIMIC-IV and MIMIC-CXR) need human-subjects training that can take several weeks, so teams planning to use them should begin in Week 1. Open datasets such as PTB-XL, MedMNIST, and TCGA are available as alternatives. Computing is provided through HiPerGator.

Materials and Supply Fees
N/A

Required Textbooks and Software

- No required textbook
- Assigned papers and lecture slides will be provided
- Software: Python, PyTorch, Hugging Face libraries, HiPerGator
- A laptop is required

Recommended Materials

- [DL] Deep Learning, MIT Press, Ian Goodfellow, Yoshua Bengio, Aaron Courville.
- [UDL] Understanding Deep Learning, MIT Press, Simon J.D. Prince.
- Stanford CS231n: Deep Learning for Computer Vision. http://cs231n.stanford.edu/
- Stanford CS224N: Natural Language Processing with Deep Learning. http://cs224n.stanford.edu/

Required Computer
Recommended Computer Specifications: https://it.ufl.edu/get-help/student-computer-recommendations/
HWCOE Computer Requirements: https://www.eng.ufl.edu/students/advising/fall-semester-checklist/computer-requirements/

Course Schedule
Week 1: Introduction: From Task-Specific Models to Foundation Models; Transformer Refresher
Week 2: Self-Supervised Representation Learning
Week 3: Foundation Models for Medical Imaging
Week 4: Vision-Language Models
Week 5: Foundation Models for EHR and Physiological Time Series
Week 6: Multimodal Fusion and Generalist Medical AI
Week 7: Adapting LLMs: Retrieval and Parameter-Efficient Fine-Tuning; Project Pitches
Week 8: Agentic Systems and Tool Use
Week 9: Generative Models and AI for Science
Week 10: Spring Break (March 13-20, no classes)

Week 11: Evaluation: Benchmarks versus Clinical Trials
Week 12: Robustness, Shortcut Learning, and Fairness
Week 13: Deployment, Human-AI Teaming, and Ambient Intelligence
Week 14: Synthesis and Open Problems; Final Project Presentations
Week 15: Final Project Presentations and Wrap-Up

Important Dates (subject to change)

January 28 *Project teams formed*
February 19 *Project proposal due*
February 23 *Project pitches; peer reviews assigned*
March 26 *Project progress report due*
April 15 and 20 *Final project presentations*
April 21 *Final project paper and repository due*

Attendance Policy, Class Expectations, and Make-Up Policy

Attendance:

- Required
- Attend office hours if having questions
- Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies. Click here to read the university attendance policies: https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/

Expectations:

- Read both assigned papers before each Thursday session and submit the reading responses on time.
- Write your own code and text. Do not copy from others (plagiarism will be checked).
- Contribute equally within your presentation, critique, and project groups.
- Ask for help if you need it (the instructor holds office hours every week).

Late Policy:

- For late project deliverables, (number of late days * 5%) will be deducted from the deliverable grade. Reading responses are not accepted after the session they cover.

Evaluation of Grades

Assignment	Total Points	Percentage of Final Grade
Paper presentations and critiques	400	40%
Reading responses and participation	200	20%
Project proposal	50	5%
Project progress report	50	5%
Final project presentation	100	10%
Final project paper and repository	200	20%
	1000	100%

Grading Policy

Percent	Grade	Grade Points
92.9 - 100	A	4.00
90.0 - 92.3	A-	3.67
86.7 - 89.9	B+	3.33
83.4 - 86.6	B	3.00
80.0 - 83.3	B-	2.67
76.7 - 79.9	C+	2.33
73.4 - 76.6	C	2.00
70.0 - 73.3	C-	1.67
66.7 - 69.9	D+	1.33
63.4 - 66.6	D	1.00
60.0 - 63.3	D-	0.67
0 - 59.9	E	0.00

Academic Policies & Resources
To support consistent and accessible communication of university-wide student resources, instructors must include this link to academic policies and campus resources: https://go.ufl.edu/syllabuspolicies. Instructor-specific guidelines for courses must accommodate these policies.

Commitment to a Positive Learning Environment
The Herbert Wertheim College of Engineering values varied perspectives and lived experiences within our community and is committed to supporting the University's core values.

If you feel like your performance in class is being impacted, please contact your instructor or any of the following:
• Your academic advisor or Graduate Coordinator
• HWCOE Human Resources, 352-392-0904, student-support-hr@eng.ufl.edu
• Pam Dickrell, Associate Dean of Student Affairs, 352-392-2177, pld@ufl.edu